

Ahmed El-Hajjar

(Legacy Software Experience Resume)

To be viewed in conjunction with primary resume at <https://resume.cdahmedeh.net>

This resume shows software development positions between 2011 and 2020. Before my transition to technical writer.

For more current projects, look at my portfolio <https://portfolio.cdahmedeh.net>

My currently active flagship project PoetWrite is a desktop application written in Java <https://github.com/cdahmedeh/PoetWrite>

Experience

Senior Software Developer and Architecture Documentation Specialist – Cisco – *Full-Time*

Jun 2020 – Oct 2020

Telecom and Networking

Kanata, ON

Technologies

Java, Go, Spring, Rally, Confluence ...

Port existing MSX from Java to Go

- MSX is an online networking system that allows Cisco to deploy routers, switches, VoIP over the cloud without requiring any additional hardware.
- For performance and efficiency reasons there was an effort to port MSX from Java (Spring Framework) to Go. I was part of that effort.
- Additional APIs were developed to make interacting with AWS simpler.

MSX Documentation

- Wrote extensive documentation about the architecture of MSX including diagrams, flow charts and UML.
- Proper software engineering practices were followed including the writing of tickets, comments, user stories, tasks, API documentation, manual and automated test cases and more.

Senior Consultant and Test Case Writer – ESDC Service Canada – *Contract*

Jan 2020 – Jun 2020

Government Public Sector

Gatineau, QC

Technologies

.NET Core, Azure DevOps, Selenium, Citrus ...

Project 1 – UI Testing Scripts for Modern ITRDS System

Develop UI Testing Scripts through Selenium framework using custom made Excel Spreadsheets
Generate C# code from custom made scripts and optimize them and fix bugs.

Project 2 – Transform Requirements to Test Scenarios

Understand requirements and changes from legacy systems to generate test scenarios.
Manually test the scenarios to ensure that all criteria pass.
Once manual testing was done, develop automated test scripts using custom tools and Selenium.

Web Developer, Server Admin and Deployment Documentation Writer – Ford Motor Company – *Contract*

Feb 2019 – Dec 2019

Automotive

Kanata, ON

Technologies

Jenkins, VirtualBox, Habitat, systemd, SLES, Spring Boot, Java, GitHub API, JIRA API, Flyway, Docker, DNS, HTTPS, Apache, nginx, Jetty, LDAP, OpenGrok ...

Project 1 – What's In

- What's In is a Spring Boot project to map JIRA tickets to GitHub changesets using their respective APIs.
- Work on persisting user sessions using database mapping done through built-in Spring Libraries
- Correct issues with use of old APIs.

Project 2 – Jenkins Enhancements

- Create Jenkins Projects to Automate Deployment of existing Internal Tools.
- Write Jenkins Plugins to enhance reliability of build projects and keeping track of state of Virtual Machines
- Setup Jenkins using custom scripts to discard build machines if builds constantly failed.

Project 3 - Dockerization of Internal Tools

- Using Docker to keep existing web projects in isolated containers.
- Automated startup and shutdown of Docker projects through systemd.
- Handle mapping of DNS, load balancers and HTTPS certificates.
- Document every aspect of the Docker setup to permit redeployment and recreation of container.
- Containerized projects included configuration and setup of Apache, nginx, Spring Boot and Jetty servers.
- Deal with LDAP authentication to limit access to these services.
- Dockerized OpenGrok while keeping source code external to the container kept secure using Linux permissions.
- An Apache file browser was Dockerized. The goal was to browse Jenkins artifacts directly reducing load on Jenkins. The file browser was secured through LDAP to only allow authorized Jenkins users to view the artifacts.

Senior Software Engineer and API Specialist – ShopLiftr – Full-Time
Marketing

Oct 2018 – Feb 2019
Ottawa, ON

Technologies

Java, REST APIs, JSON, Spring, Groovy, SQL, Hibernate, Gradle, Maven, Amazon AWS, SageMaker, Amazon ML, Amazon EMR, Amazon Datapipeline, CloudWatch, bash, git, Microsoft SQL Server, DynamoDB, Bitbucket, ...

Admin Portal – Machine Learning

- Run of migrating legacy Amazon ML system to SageMaker for more accurate predictions on product prices and to limit product price entry errors from data teams.
- Explored different algorithms like XGBoost and various other linear and categorical ML algorithms to find optimal prediction results.
- Improved results dramatically of prediction system. RMSE was brought down to less than 1 detecting common errors in price input by the data entry teams.

AWS Infrastructure - Maintenance

- Basic maintenance and fixes for issues on Amazon AWS.
- Communicate with Amazon on bugs found in their service.
- Optimizations to the architecture and working on simplifying it.
- 24/7 monitoring of systems on an on-call basis.

Full-Stack Development

- Fix and maintain application using Spring backend written in Groovy with Angular 6 frontend.
- Reduce performance issues due to limitations in Groovy language.
- Enhance branding and look and feel through Photoshop.

Swagger Migration

- Migrate all API documentation to Swagger.
- Allowed team to test APIs without using any external tools.

Documentation Enhancement

- Set protocols for detailed JIRA tickets including API functionality and error reporting and estimates.
- Create and update existing documentation for undocumented parts of the stack.

Technologies

Java, REST APIs, JSON, Spring, Spring Boot, SQL, Postgres, H2, Hibernate, JUnit, jMock, Mockito, Apache Commons, Guava, Gradle, Maven, Django, Blockchain, SDR, Estimote, Amazon AWS, Firebase, APNs, CloudWatch, LogEntries, Microsoft Azure, Portainer, Docker, Docker Swarm, TeamCity, bash, git, BitBucket, Linux, Android, Google Maps API, Microsoft Cognitive Services, OCR...

LWAYVE – Product Support

- LWAYVE is an innovative SDK that integrates into other Android and iOS mobile applications allowing audio streaming by determining what the user wants to hear based on time, GPS location, proximity to Bluetooth beacons and even role within an organization. In other words, LWAYVE allows the user to hear the right content at the right time.
- Customers included NASCAR, Osheaga Music Festival, Halifax Pop Explosion and many others.
- Developed documentation and organized meetings to improve existing software and suggest new ideas that helped the company stay innovative and flexible. As a result, sales have more pitches to sell the software to more customers as the software had more features and was more stable.
- Kept a constant stream of communication with product owners and managers enhancing understanding of user stories, specifications and product goals.
- Create new designs and architectures to reduce technical debt and enrich software functionality.
- Followed various agile principles such as SCRUMs, Pair Programming, Code Reviews, Grooming Sessions and KANBAN to maximize developer productivity and release products in a timely fashion.

LWAYVE – Backend

- I was the sole owner of the backend for LWAYVE and the only developer. I was responsible for all bugfixes, feature implementation, user stories, documentation and demos.
- Created RESTful API web service using Spring Boot and JHipster. API was used by all parts of application including iOS SDK, Android SDK, Web Portal, RedShift Analytics and GSheet Playlist Generator.
- Developed a Push Notification System using Firebase Google Cloud Messaging and APNs (Apple Push Notification Service) to send instant notifications to users.
- Built an efficient audio stream storage system using Amazon S3 Cloud Storage. It allowed both mobile devices to send recordings and the web portal to upload MP3 files.
- Playlist JSON was designed with an upgradable schema with automatic versioning through the Jackson JSON serializer and versioning module. It reduced efforts to add features to playlisting functionality and allowed backwards and forwards compatibility with the mobile SDKs.
- Automatic documentation of APIs and web testing portal implemented with Spring Annotations and Swagger UI.
- Optimized playlist retrieval API with caching using Hazelcast. This allowed millions of requests to be done per minute without making any database calls to Postgres.
- Backend ran in micro-services architecture using Eureka and Zuul in a Docker Swarm.
- Assisted QA in developing test cases for test automation thus reducing regressions.
- The backend contained unit tests through JUnit, JMock and Mockito using an in-memory H2 database. This reduced regressions and we had code coverage of 90%. I would refuse code reviews if no manual testing was done.
- While almost all database tasks were handled via Hibernate, I sometimes implemented custom queries for optimizing frequently used APIs such as playlist retrieval.

LWAYVE – Documentation

- Composed User Stories for the product team in order to assist in defining new features and fixes along with acceptance criteria.
- Extremely thorough API documentation with regular usage, incorrect usage, success messages and failure messages specifically designed for mobile and web developers.
- Proper Software Engineering practices through detailed tickets, code reviews, comments and more.
- Marketing and Sales documents to increasing area of sales pitches for sales team.
- Architecture documentation including diagrams for alternate designs for the software.

LWAYVE – DevOps

- Maintain deployment tasks via TeamCity with automated bash scripts reducing deployment times to several minutes. Downtime between updates lasted only 2-3 minutes though the mobile applications remained functional during the downtime.
- Preserve logs across sessions with Amazon CloudWatch and LogEntries allowing easy debugging of staging servers or production server errors.
- Create configuration files and deployment scripts including Docker Compose files and store them in git repositories to allow code reviews and versioning.
- The micro-services could be scaled automatically using AWS Auto Scaling Groups.
- Deployed micro-services to both Microsoft Azure and Amazon AWS. JMeter was used to benchmark any performance differences.
- Documented every failure of production in Confluence including time, failure mode and resolution. This allowed the team to resolve persistent issues without having to discover the cause again.

LWAYVE – Release Support

- Handled all release branching on git include merges to master and tagging.
- Implemented automatic merging functionality in BitBucket that merges code into downstream branches. This simplified how code was merged into the various repositories without manual intervention improving developer productivity and prevented accidental merges into wrong branches.

LWAYVE – Android

- Helped in the creation of Native Android SDK to allow clients to integrate our technology into their mobile applications.
- Designer of Decision Tree Library which was shared by the backend and Android SDK. A dependency-free design allowed the Android SDK to maintain high performance on less powerful devices.
- Alleviated workload on main Android developers with bug fixes and code reviews.

Design and Concept Team – Sales

- The Design and Concept is a Research and Development lab at Lixar that explored experimental technologies to determine emerging tech and possibly the future of Lixar.
- The Vision and Implementation had to be implemented in short time frames that lasted less than a few days.
- Demos to sales were made to demonstrate how the product could be sold.
- Provide support for sales when required. Some projects would eventually become production quality and sold to clients.

Design and Concept Team – Blockchain

- Was the pioneer of blockchain at Lixar. Few understood the technology, so I took the task of making presentations to explain it in very simple terms.
- Created a proof-of-concept using Multichain and Django (Python) that would allow multiple organizations to share files instantly and verify their authenticity.
- The concept led to a sale to Enform, a certification validation organization and the blockchain concept was also used to remunerate artists for having their tracks played on LWAYVE.

Design and Concept Team – Proximity Projects

Developed and tweaked proximity sensors with Bluetooth beacons improving performance of ProxSee product. ProxSee is software that tracks user locations without draining battery with low-energy Bluetooth beacons and GPS.

Implement Wi-Fi sniffers to detect nearby smartphones without needing additional software.

Software defined radios tuned to cell phone tower frequency to detect location with triangulation. Became a prototype functionality for ProxSee.

Design and Concept Team – Face Recognition

- Using the Microsoft Cognitive Services API, leveraged face detection and recognition with facial parameters to measure analytics.
- Combined motion detection and hardware facial recognition for initial scan for enhancing performance.
- Was used successfully as a PoC in a pet store, Pet Valu, to implement a dashboard that demonstrated returning customers, age, gender, time of entry and exit, and other parameters. They are looking to implement similar technologies for detailed analytics.

Android Mobile App – MVCS Crash Detection

- Designed a PoC for detecting vehicles crashes through accelerometers and microphone levels.
- Updated branding of application to match new company logos with image editing tools.

SRED Project – FlyScan

- Designed tool for finding original document by scanning or photographing a loose page.
- Used OCR for detection of words in the document.
- Enhanced text search performance by reducing search space and heuristics for Lowenstein distance and common English words.
- Explored Machine Learning, Neural Networks and Map-Reduce as alternatives to OCR.
- Wrote entire SRED Document in order to display the innovative nature of the product along with all the components and research efforts and procedure.

Mentoring

- Helped other developers (intermediate and junior) to get up to speed on new technologies.
- Mentor junior developers and help with career progression.
- General go-to-person for questions on software stacks on various platforms include the JDK.

Software Engineer – NorthStar Utilities Solutions – Full-Time *Utilities Sector*

Dec 2013 – Oct 2016
Ottawa, ON

Technologies

Java, REST APIs, XML, XML-DTD, Java EE, EJB, JAX-RS, Swing, JSP, JSF, PrimeFaces, SQL, Microsoft SQL Server, Informix SQL, Oracle SQL, JDBC, Hibernate, Windows Server, Linux, Unix, HP QTP/UFT, Ant, Tomcat, JBoss, WildFly, Bamboo...

NorthStar Customer Information System

- Implement new features and bug fixes as a full-stack developer in all layers of application whether it was database on JDBC, backend running in Java EE, desktop via Swing and SWT or Web UI in JSP and JSF.
- Implemented a database caching feature that would retain results from SQL queries.
- Redesigned desktop UI based on Swing and SWT to have a more modernized appearance.
- Wrote multi-database compatible queries that worked in Informix, Oracle and Microsoft SQL Server in the backend.
- Was responsible for Customer Information System, Billing, Financing, Automation Platform and Customer Portal Components.

NorthStar Automation Platform

- Maintained Automation Platform which allowed clients to automate certain tasks via a graphical environment.
- Created a SQL Query converter using ANTLR that would parse queries and convert them to queries compatible with Informix, Oracle and SQL Server. Missing functionalities would be handled via stored procedures.

NorthStar Releases

- Handled software releases, packaging and deployment. I automated much of the tasks via Windows Shell Scripts, Ant and PowerShell

Insight Project

- Insight was an experimental project that would allow users to create dashboards based on Web APIs, web page scraping, CSV files and database connections.
- Stack was created with a modern backend and frontend using Java EE, JSF and PrimeFaces.
- Aspects of the stack were transferred to the CIS (Customer Information System).

Initiatives for Architectural Improvements

- Heavy focus on architecture and code quality. Hosted meetings with superiors to discuss road-plan and include refactoring as part of future development of the software. Included detailed project plan, estimates, tickets and possible burn-down charts.
- Convinced Software Architect to have me develop a rewrite for the build scripts for the main CIS project. I rebuild the Ant build script reducing both code size by several factors and improving build time from 30 minutes to 45 seconds.
- Converted ancient use of Tomcat web server to JBoss/Wildfly vastly improving runtime of CIS.

NorthStar QA Initiatives

- Encourage developers to write test cases before QA did to ensure that QA would spend less time testing features and would require less documentation.
- Restored broken HP QTP/HTP instance which allowed QA to finally re-run their automated tests after several months of downtime.

Environments Administrator

- Maintained all R&D server infrastructure ensuring their constant functioning and writing scripts for automating tasks using batch files and PowerShell.
- Handled all database backups including nightly deployment of backups for staging servers using SQL Server Administrator.
- Deployed and provisioned environments for QA as they needed them.
- Monitor servers through SSH and RDP and respond to internal needs based on importance of failures and their impact on the various teams.
- Maintained Bamboo for automatic builds and created new build projects. Reduced workload on support since they didn't need to build the project manually or get a copy from developers.

Mentoring

- Helped other developers get up to speed on new technologies.
- General go-to-person for difficult questions on JDK.

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Software Developer – Teldio – Full-Time
Telecom

May 2013 – Aug 2013
Kanata, ON

Technologies

- Java, Swing, Ruby on Rails, Asterisk, PBX, Twitter Bootstrap, HTML, CSS, AJAX, JavaScript...

Teldio Portal

- Used Ruby on Rails for implementing bug fixes and new features.
- Extensively worked with Asterisk, PBX and SIP Trunks for managing radio-to-phone communications.
- Developed emergency alerting system API for radio and telephones through SIP.
- Modify InnoSetup scripts when software dependencies changed.

Internal Tools

- Developed internal tools to increase productivity of employees using Ruby on Rails. Tools included automated releases to S3 that would compile from source. Twitter Bootstrap was used for making the internal tools look more pleasing.

Mentoring

- Mentor a senior developer who was unfamiliar with Swing.

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Technologies

Java, Java EE, JSP, JSF, GIMP, Jenkins...

MediaVantage Portal

- Followed Agile Software Principles as a software engineering methodology.
- Used Test Driven Development practices to prevent regressions and for validation using JUnit.
- Managed and implemented VMs using VirtualBox on various Linux platforms for testing web services.
- Maintained and corrected bugs for advanced web services written using JSP, JSF, Java and other web service technologies.
- Wrote documentation for web services deployment that use Tomcat and Pentaho software technologies.
- Redesigned UI elements by hand using GIMP to modernize appearance.
- Created modules for exporting reports in native Microsoft Open XML format.
- Modifying Jenkins scripts due to new features.

Technologies

Java, Swing, JDBC, Ant, Unix, Informix, 4GL...

Customer Information System – Sustaining Team

- Member of the Sustaining Team focused on bug fixes and new features implementation written in Java on a Java EE platform.
- Wrote software patches to implement new features based on client requirements.
- Modified legacy Informix-4GL code on classic Alpha DEC Unix environment.
- Used version control system CVS to submit code changes.
- Familiarized myself with code review processes and pair programming.
- Prepared presentations for coworkers and clients for various purposes.

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